SONAL SHAW

B.Sc. Data Science and Statistics | CHRIST (Deemed to be University), Bengaluru
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PROFESSIONAL SUMMARY

Third-year B.Sc. Data Science and Statistics student with hands-on experience in data analysis, dashboard design, and applied machine learning. Skilled in Power BI, Tableau, SQL, Python, and R, with a strong statistical foundation and proven ability to build full-stack and NLP-based projects from raw data to actionable insight. Seeking a Data Analyst job to apply analytical and technical skills to real business problems.

EDUCATION

CHRIST (Deemed to be University), Central Campus, Bengaluru *Present*
Bachelor of Science in Data Science and Statistics | GPA: 3.7/4

Calcutta Girl's High School, Kolkata *March 2024*
XII, ISC | 93%

Calcutta Girl's High School, Kolkata *March 2022*
X, ICSE | 94%

PROJECTS

A Multi-Agent AI Framework for Explainable Credit-Risk Assessment | *Python, LangChain, XGBoost, SHAP*

- Authored a research paper, currently **under review for publication**, proposing a five-agent LLM framework (LangChain, Llama 3.2) for explainable credit-risk assessment on merged Indian Bank and CIBIL data (51,336 records).
- Designed agents for data reconciliation, risk prediction, policy checking, explanation and orchestration, using deterministic Python tools so the LLM reports results rather than generating numbers.
- Compared XGBoost, Decision Tree and Random Forest; XGBoost performed best at 99.48% accuracy with credit score and 80.54% without it, showing risk classes depend mainly on one variable.
- Used SHAP for feature-level explanations; the Explainability Agent matched SHAP's top-5 features with 96% overlap and 90% direction agreement across 20 customers.

Smart Healthcare Website | *Python, Django, HTML, CSS, SQLite*

- Built a full-stack healthcare management web application in Django, implementing user login/signup, appointment booking, doctor and medical records pages, and a teleconsultation module.
- Designed 7+ integrated frontend pages (home, doctors, appointments, records, login, signup, teleconsult) with HTML/CSS templates connected to a Django backend.
- Structured backend logic using Django models, views, URL routing, and SQLite migrations, demonstrating end-to-end full-stack development and healthcare workflow functionality.
- Deployed complete source code to GitHub, organizing project, app, template, and static asset directories for maintainability.

GitHub: https://github.com/Sonal-Shaw/smart-healthcare-website

Fake Review Detection – Deceptive Hotel Review Classification | *Python, NLP, Scikit-learn*

- Built an NLP/ML pipeline to classify hotel reviews as genuine or deceptive using the Deceptive Opinion Spam dataset.
- Cleaned and preprocessed text data; engineered features via TF-IDF vectorization, word/character counts, average word length, exclamation marks, and uppercase ratio.
- Trained and compared 6 classification models (KNN, Decision Tree, Random Forest, Naïve Bayes, SVM, Logistic Regression) evaluated on accuracy, precision, recall, and F1-score.
- Identified Naïve Bayes as the top performer at 93.5% accuracy; visualized and interpreted results to support model comparison and selection.

CERTIFICATIONS

- Data Visualization in Power BI — DataCamp, Sep 2024
- Data Visualisation: Empowering Business with Effective Insights — Tata (Forage), Oct 2026

SKILLS

Technical: Microsoft Excel, SQL, Tableau, Power BI, R Programming, Python, Java, Canva
Soft Skills: Critical Thinking, Analytical Thinking, Time Management
Languages: Hindi, English, Bengali